In other words, any student in the undergraduate program can evaluate their

- Eligibility to enroll in the certificate and to plan their future courses
- Progress through the certificate program as they complete courses
- Certificate fulfillment status, including the GPA requirement, on completion of all the course requirements

> **NOTE** Students in this situation do not violate Family Educational Rights and Privacy Act (FERPA) law or any other regulations because they are dealing with only their own grades and do not access the central student database.

Currently, students have to schedule a one-on-one appointment with an adviser to verify their fulfillment status for the certificate. This model is not a scalable solution if thousands of students want to enroll in the certificate program. This DCC-Dapp will save time for the students as well as the advisers, and streamline the certification process for some. More important, the recording on the blockchain of transactions initiated by interested stakeholders will be a valuable resource and institutional data for future analytics for course planning, advisement, and resource planning.

11.3.1 Context for the DCC application

This problem refers to a specific context of which I have firsthand knowledge, so I am fortunate to have some specific details. Figure 11.2 gives an overview of the problem under consideration. The certificate includes four categories of courses and a minimum GPA in these courses. Any stakeholder, student or adviser, can register by using a decentralized identity the first